



10 CFR 50.73

LG-19-101

October 24, 2019

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2019-001-00, Multiple Trains of RHRSW Inoperable for Greater than Time Allowed by Technical Specifications due to Pipe Corrosion

Enclosed is a Licensee Event Report (LER) that addresses multiple trains of Residual Heat Removal Service Water (RHRSW) being inoperable for greater than allowed time by Technical Specifications due to piping corrosion. This condition is reportable for a condition prohibited by Technical Specifications and a common cause inoperability of independent trains.

This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(vii).

There are no commitments contained in this letter.

If you have any questions, please contact Robert B. Dickinson at (610) 718-3400.

Respectfully,

A handwritten signature in black ink, appearing to read "F. Sturniolo", written over a horizontal line.

Frank Sturniolo
Vice President – Limerick Generating Station
Exelon Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station



Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (7-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Information.Resource@nrc.gov and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
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NARRATIVE**Unit Condition Prior to the Event**

Limerick Generating Station (LGS) Units 1 and 2 were both operating in Operational Condition (OPCON) 1 at 100 percent power at the time of the event. There were no other structures, systems, or components inoperable at the time of the event that contributed to the event.

Description of the Event

On 7/11/19 a pin hole leak was identified on the 18-inch diameter distribution piping of the C Spray Network while the spray network was in operation for Spray Pond cooling. Subsequent NDE inspections identified wall thinning over a large arc of the piping's circumference; approximately 34 feet of the 18-inch diameter piping required replacement. With this condition present, the spray network was successfully operating and performing its intended function.

The Spray Pond [BS] serves both units at Limerick Generating Station (LGS) and is designed to supply and cool the safety-related cooling water for the Emergency Service Water (ESW) [CC] and the Residual Heat Removal Service Water (RHRSW) [CC] systems. The Spray Pond contains four spray networks. Each network contains 240 spray nozzles with all the piping and spray nozzles above the water level supported by concrete pillars. Only one spray network is required for the safe shutdown of one unit and any two networks provide sufficient cooling to safely shut down both units.

During the piping preparation for Non-Destructive Examination (NDE) additional through-wall piping pin holes were identified. Due to the number and tight physical spacing between defects, ASME Code Case N-513 could not be applied to this condition to provide an interim operability justification. Subsequently, the C RHRSW spray pond spray network was isolated and declared inoperable on 07/13/19 and Limerick entered an emergent 30-day dual unit Tech Spec LCO. The degraded condition on the east side of the supply manifold was corrected and the LCO was exited on 07/30/19. An extent of condition plan was developed to perform NDE on the remaining three spray pond spray networks.

On 08/26/19 the A Spray Network was removed from service and the LCO was entered for an extent of condition inspection. On 08/27/19 during piping preparation for NDE, pin holes were identified on the east side of the 18-inch distribution piping at the same location as the C Spray Network piping. The piping corrosion damage was identified on the 18-inch distribution piping on both the east and west sides. The piping wall thickness readings on the east side were below the minimum required thickness; however, the piping wall thickness readings on the west side were above the required minimum thickness. This condition was determined to be reportable for a condition prohibited by Technical Specifications and common cause inoperability of independent trains. The east side 18-inch distribution piping was replaced and the LCO was exited on 09/07/19.

On 09/10/19 the B Spray Network was removed from service and the LCO was entered for the extent of condition inspection. No pin holes were identified; however, NDE inspection identified similar corrosion damage on the 18-inch distribution piping on both the east and west sides. The piping wall thickness readings on the east side were below the minimum required thickness; however, the piping wall thickness readings on the west side were above the required minimum thickness. The east and west side 18-inch distribution piping was replaced and the LCO was exited on 09/21/19.

On 09/23/19 the D Spray Network was removed from service and the LCO was entered for the extent of condition inspection. No pin holes were identified; however, NDE inspection identified similar corrosion damage on the 18-inch distribution piping on both the east and west sides. The piping wall thickness readings on the east side were below the

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minimum required thickness; however, the piping wall thickness readings on the west side were above the required minimum thickness. The east and west side 18-inch distribution piping was replaced and the LCO was exited on 10/03/19.

Analysis of the Event

Each spray pond spray network is fed water by a single 30-inch diameter pipe. A tee divides the flow into two (east and west side) 18-inch diameter sections. This manifold distributes all the water to the spray nozzles by seven individual 10-inch diameter piping headers equipped with spray nozzle trees, that contain four spray nozzles per tree. The NDE identified degradation in the 18-inch piping; no significant degradation was identified in the 30-inch or 10-inch piping. The A and C Spray Networks exhibited specifically oriented, localized wall thinning and through-wall pin holes on the 18-inch distribution piping while the B and D Spray Networks exhibited specifically oriented, localized wall thinning, but no pinholes.

The apparent cause of the pipe corrosion damage observed on the 18-inch distribution piping of each spray pond spray network was flow-induced film removal accompanied by high humidity electrochemical atmospheric corrosion. The required standby alignment for the spray pond spray networks results in the piping normally being drained and open to the atmosphere. When the system is placed in service, the air from the network piping and the subsequent flow oscillations, driven by the piping geometry, results in removal of any scale and corrosion products formed during the time the piping was out of service. When the system is not in service the newly cleaned pipe surface is exposed to the atmosphere in high humidity conditions. The corrosion footprint in the areas of excessive pipe corrosion and through-wall defects matches the corrosion footprint of excessive atmospheric corrosion.

NDE inspection of spray pond spray network piping has previously been performed on the A and D Spray Networks at thirteen locations on each spray network. The intent was that the inspection results would be representative of all spray network piping. NDE inspection was last performed in June 2011. Two of the thirteen inspection locations were located on 18-inch distribution piping and the NDE results for these inspections identified reasonably healthy piping with no evidence of the excessive focused wall thinning. However, only localized scanning (spot readings) at the top, bottom and sides of piping was performed.

LGS has experienced pin hole leaks in other RHRSW and ESW piping and is proactively replacing portions of the piping throughout the plant. The previous efforts to prioritize RHRSW and ESW pipe replacements determined the Spray Network piping was low risk. This was based on limited NDE data that indicated no significant corrosion issues and a lack of understanding of the corrosion mechanism.

Safety Consequences

Although the piping was degraded, the spray networks and the ultimate heat sink were considered functional. Routine operation of the spray networks demonstrates that the spray networks remained capable of performing their function under non-faulted conditions. The faulted conditions described in the UFSAR (seismic events and tornado (excluding missiles)) are not postulated to occur coincident with a design basis accident.

The as-found NDE data for the A Spray Network was used to perform a realistic structural integrity assessment of the piping. This assessment concluded that none of the stresses resulting from the faulted events exceeded the ASME code allowable stress values. This evaluation supports the judgment that the piping would remain intact and not result in a

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gross failure that would challenge the heat removal function of the spray pond spray network for a seismic event or tornado (excluding missiles).

Paper tracings and photographs of the through wall flaws of the A Spray Network piping were used to perform an assessment to determine if the flaws could, under faulted loading conditions, connect and result in localized material yielding and formation of hairline type slits that would cause additional external leakage. The conclusion from this assessment was this could occur in a few locations, but the size of any newly formed hairline slits would be tolerable and not result in a significant amount of bypass leakage that would adversely impact the heat removal function of a spray network. Slightly less than 140 gallons per minute (gpm) was the computed bypass leakage which is well within analyzed limits.

Although not expected or anticipated, the consequences the postulated loss of some or all the Spray Pond spray networks due to tornado missiles is specifically addressed in the LGS UFSAR. The Limerick Spray Pond spray networks have no protection against tornado generated tornado missiles. The UFSAR and existing plant special event procedures provide remediating actions to successfully address and mitigate loss of spray networks. These actions could also be utilized as a mitigation strategy for any of the postulated external events given the long-time frames involved in a scenario with degraded spray pond spray networks.

Cause of the Event

The cause of the pipe corrosion damage observed on the 18-inch distribution piping of each spray pond spray network was flow-induced film removal accompanied by high humidity electrochemical atmospheric corrosion.

The condition was not identified prior to the piping becoming inoperable because there was a lack of understanding of this failure mechanism.

Corrective Actions

- All inoperable 18-inch diameter distribution piping was replaced.
- Complete the Extent of Condition inspection on the C Spray Network.
- Establish NDE inspection and proactive pipe replacement strategy to ensure long term reliability of the above ground spray pond spray network piping.
- Enhance current NDE inspection strategy for ESW and RHRSW piping to proactively inspect pipe segments with limited NDE inspection history.

Previous Similar Occurrences

Prior to the C Spray Network piping through wall leak identified on 07/11/19, all RHRSW System and ESW piping through-wall leaks and minimum wall thickness issues have occurred on piping that is normally filled with water. LGS has extensive operating experience dating back to early plant life indicating that corrosion of ESW and RHRSW piping that is normally filled with water is characterized as general corrosion with more severe localized attacks (pitting) resulting from under deposit corrosion influenced by micro-biologically induced corrosion (MIC).



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The corrosion mechanism that resulted in the specifically oriented localized corrosion damage of the 18-inch spray network distribution piping is different than any safety related raw water piping corrosion damage previously observed at LGS.

System: BS
Component: PSP
Component Number: N/A
Component Name: Spray Pond Spray Network
Manufacturer: N/A